

NeuroPilot

A premium multi-agent operating system for ML/DL-driven neuroengineering and neuroscience labs. Move from scattered tooling to a governed, end-to-end research execution pipeline.

- NEUROENGINEERING
- COMPUTATIONAL NEUROSCIENCE
- ML/DL EXPERIMENT OPS
- CLAIM & CITATION AUDIT
- PI GOVERNANCE LAYER

Product Positioning

Before Research operations are split across chat logs, notebooks, ad-hoc scripts, and untracked decisions. Teams cannot reliably connect claims to evidence, and context gets lost between iterations.	With NeuroPilot Every stage is orchestrated through a role-based multi-agent stack with explicit approvals, budget controls, execution telemetry, and reproducible artifact generation for publication workflows.
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Agent Organization

PI STRATEGY · SIGN-OFF Owns prioritization, budget ceilings, and final approval checkpoints.	Ideator + Lit Scout HYPOTHESIS · PRIOR ART Builds falsifiable hypotheses and validates novelty windows.	Experimenter + Engineer PLAN · BUILD · RUN Converts hypotheses into reproducible ML/DL experiments and GPU runs.
Critic INTEGRITY AUDIT Performs adversarial checks on claims, citations, and result consistency.	Writer PAPER PIPELINE Transforms validated evidence into paper, poster, and talk artifacts.	Worker + Mentor OPS · STRATEGIC REVIEW Keeps execution unblocked and injects external strategic feedback loops.

Execution Flywheel

01 Scope Thesis Refine a broad direction into a measurable neuro-AI hypothesis.	02 Map Prior Art Establish baselines, novelty boundaries, and evaluation targets.	03 Launch Experiments Run seeded experiments with explicit configs and run metadata.	04 Audit Claims Force evidence-traceability and reject unsupported statements.	05 Publish Assets Generate paper-ready outputs after PI gate approval.
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Why It Matters

OPERATIONAL RELIABILITY Always-On Execution Heartbeat-driven automation keeps progress running around the clock.	COST GOVERNANCE Hard Budget Stops Caps and policy gates prevent runaway token spend and drift.	SCIENTIFIC TRUST Evidence-Linked Claims Every major claim is auditable against concrete experiment artifacts.	RESEARCH VELOCITY Shorter Paper Cycle Integrated planning, execution, and writing compress time-to-submission.
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